

Parallel & Distributed Computing



COMPUTER CLUSTERS



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Lesson from Quran

2



Lecture Outline

3

- Introduction to Computer Clusters
- Why Do We Need Clusters?
- Cluster Architecture Overview
- Cluster vs Grid Computing
- Types and Applications of Clusters
- Cluster Interconnection Networks
- Software Stack in Cluster Systems
- Load Balancing and Fault Tolerance
- Summary



Introduction to Computer Clusters

4

- A **computer cluster** is a group of independent, interconnected computers (nodes) that work together as a unified computing resource.
- Nodes communicate through a **high-speed local network (LAN)** to perform large-scale computation or service delivery.
- From a user's perspective, the entire cluster acts as a **single system image (SSI)**.
- Goal: Enhance **performance, scalability, availability, and fault tolerance**.
- **Example:**
 - ▣ Google Search Engine uses thousands of clustered servers to index and respond to billions of queries per day.



Why Do We Need Clusters?

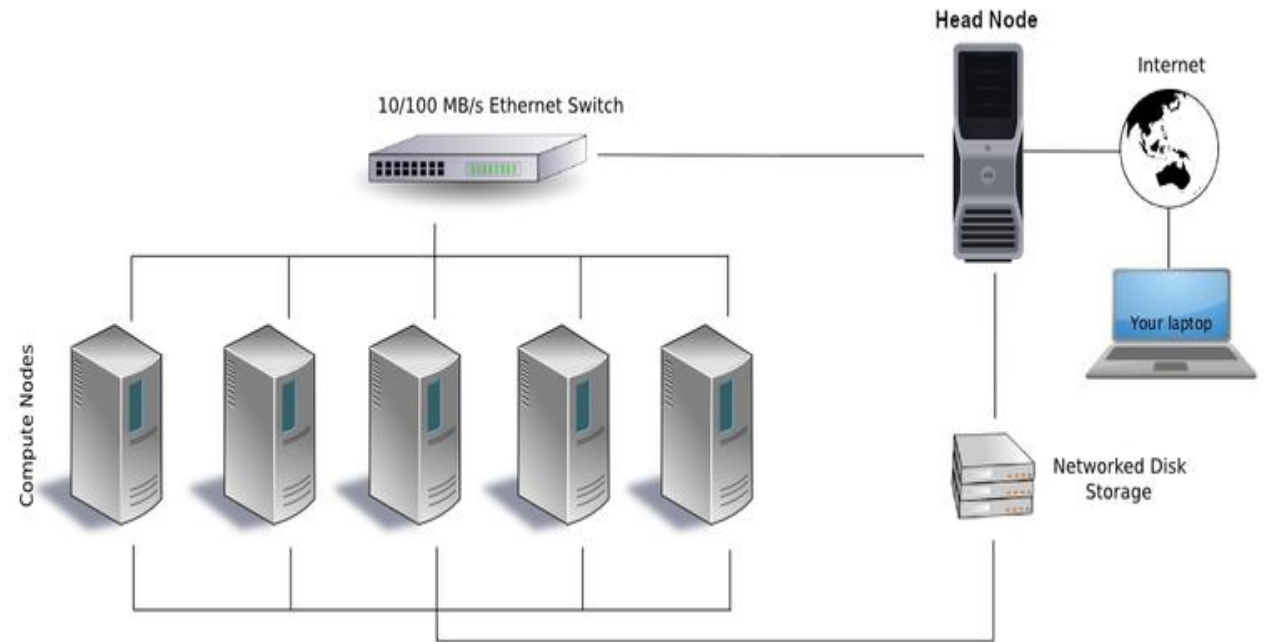
5

- ❑ **Scalability:** Handle growing workloads without replacing the entire system.
- ❑ **Performance:** Parallel execution across multiple nodes reduces computation time.
- ❑ **Cost-Effectiveness:** Uses commodity hardware instead of expensive supercomputers.
- ❑ **Reliability:** Failure of one node doesn't bring down the system.
- ❑ **Flexibility:** Nodes can be added or removed dynamically.
- ❑ **Reading Assignment:** Pacheco, Ch. 3 – *Clusters and Networked Systems*

Cluster Architecture Overview

6

- ❑ A typical cluster includes:
- ❑ **Head Node (Master Node):**
 - ❑ Manages resources, scheduling, and job distribution.
 - ❑ Controls login and data access.
- ❑ **Compute Nodes:**
 - ❑ Perform assigned computation tasks.
 - ❑ Typically identical in configuration.
- ❑ **Interconnection Network:**
 - ❑ Provides high-speed communication (Gigabit Ethernet, InfiniBand).
- ❑ **Shared Storage:**
 - ❑ Accessible by all nodes (via NFS, Lustre, or GlusterFS).



Computer Clusters

7

- Unlike **grid computers** (where each node set to perform a different task/ application), computer clusters have each node set to perform the **same task**, controlled and scheduled by software.
- In most circumstances, all of the nodes use the same hardware and the same operating system, although in some setups (e.g. using **Open Source Cluster Application Resources** (OSCAR)), different operating systems can be used on each computer, or different hardware.
- OSCAR is a Linux-based software installation for high-performance cluster computing.

Computer Clusters

8

Cluster Computing involves connecting two or more **homogeneous** computers to work as a unified system. This approach is commonly used for tasks that **require** high computational power and reliability. Here are the key characteristics:

Advantages of Cluster Computing

- ❑ **Homogeneity:** Since the computers in a cluster are of the same type, they are easier to manage and maintain.
- ❑ **High Performance:** Clusters can offer significant computational power and speed by pooling resources.
- ❑ **Reliability:** Clusters can provide redundancy and fault tolerance if one node fails.

Disadvantages of Cluster Computing

- ❑ **Limited Scalability:** Clusters are typically limited to a single location, which can restrict their scalability.
- ❑ **Cost:** The requirement for homogeneous hardware can be costly.

Grid Computing

9

Grid Computing refers to a network of computers, which can be either **homogeneous or heterogeneous**, working together across **different locations** to perform complex tasks. It leverages the unused processing power of multiple machines to achieve its goals.

Advantages of Grid Computing

Flexibility: Grid Computing can integrate a variety of hardware and operating systems.

Scalability: It can harness resources from a global network, allowing for greater scalability.

Cost Efficiency: By utilizing existing unused resources, Grid Computing can be more cost-effective.

Disadvantages of Grid Computing

Complexity: Managing and coordinating a grid of diverse machines can be complex.

Performance Variability: The performance might vary due to the diversity of the participating machines and network latency.

Cluster vs Grid Computing

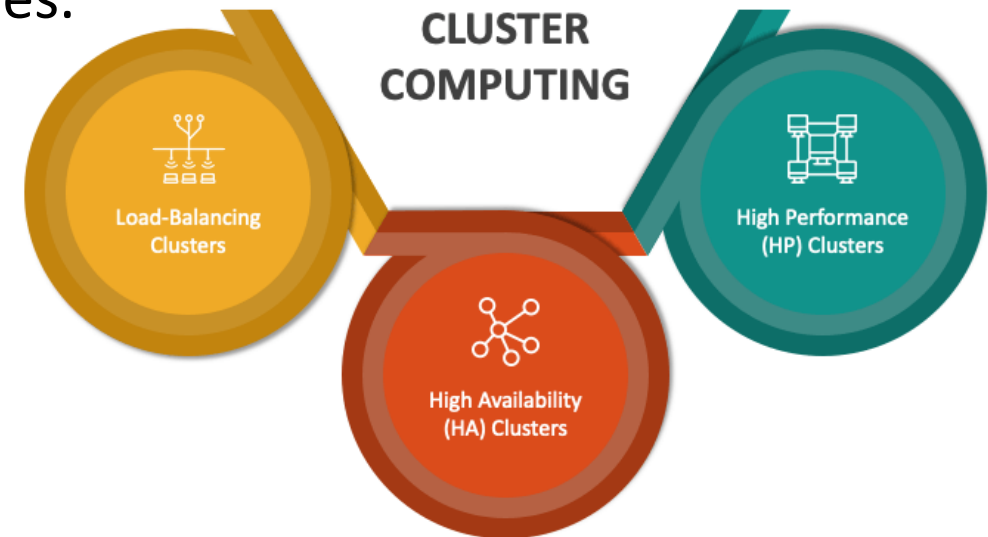
10

Aspect	Cluster	Grid	Cloud
Ownership	Single organization	Multiple organizations	Third-party providers
Network	LAN	WAN	Internet
Resource Management	Centralized	Decentralized	Virtualized
Example	Beowulf, Rocks	SETI@Home	AWS, GCP, Azure
Use Case	Scientific computing	Volunteer computation	On-demand elastic computing

Types of Clusters

11

- ❑ **High-Performance Computing (HPC) Clusters:**
 - ❑ Focus on computation-intensive tasks (e.g., scientific simulations, weather models).
 - ❑ Example: NASA's Beowulf clusters.
- ❑ **Load Balancing Clusters:**
 - ❑ Distribute incoming requests evenly across nodes.
 - ❑ Used in web servers or cloud backends.
- ❑ **High-Availability (HA) Clusters:**
 - ❑ Provide redundancy and failover support.
 - ❑ Example: Database clusters.
- ❑ **Hybrid Clusters:**
 - ❑ Combine HPC and HA capabilities.



Applications of Cluster Computing

12



Weather modeling



Image processing & electromagnetics



Solving complex computational problems



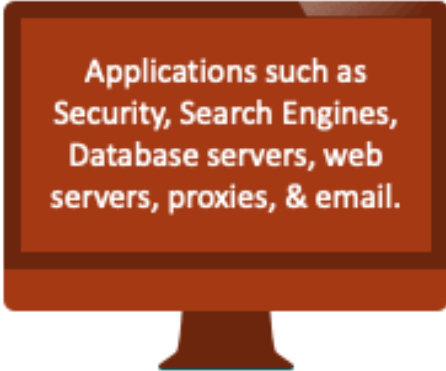
Vehicle breakdown & nuclear simulations



Astrophysics, aerodynamics, & data mining applications



Forecast the occurrence of earthquakes or tornadoes.

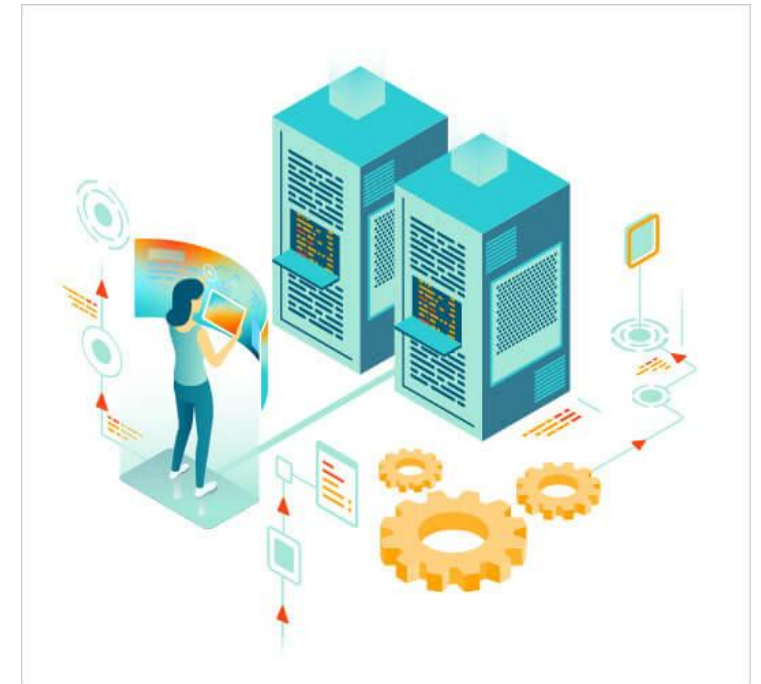


Applications such as Security, Search Engines, Database servers, web servers, proxies, & email.

Beowulf Cluster Example

13

- ❑ **Beowulf Cluster** = collection of commodity PCs running Linux connected via Ethernet.
- ❑ **Software Stack:** Linux + MPI + PVM + OpenMP.
- ❑ **Advantages:**
 - ▣ Low-cost parallel computing environment.
 - ▣ Open-source ecosystem.
 - ▣ Scalable for academic and research workloads.
- ❑ **Example Setup:**
 - ▣ 16 nodes $\times$ 8 cores each = 128-core cluster.
 - ▣ Peak performance $\approx$ 1 TFLOP at $<10\%$ cost of supercomputer.

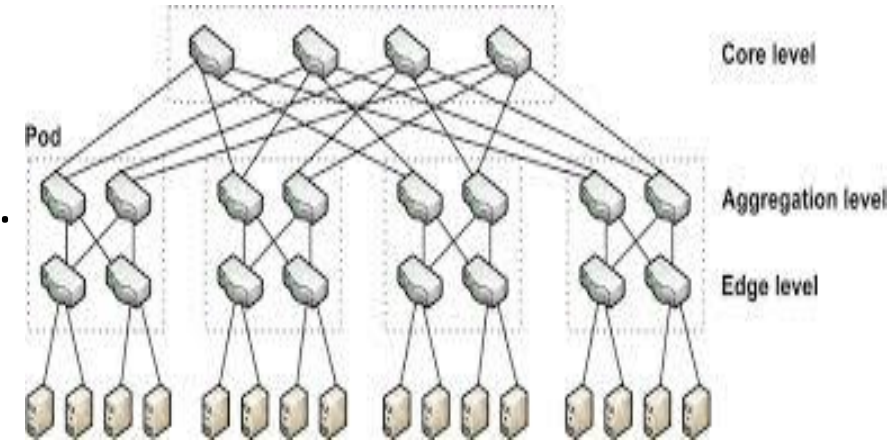


Cluster Interconnection Networks

14

- Communication backbone determines performance.
- **Topologies:**
 - ▣ **Star:** Simple but single point of failure.
 - ▣ **Ring:** Easy to implement; higher latency for distant nodes.
 - ▣ **Mesh:** High fault tolerance but complex routing.
 - ▣ **Fat-tree / Torus:** Common in HPC clusters.
- **Technologies:**
 - ▣ **Gigabit Ethernet:** Cheap, moderate latency (30–100 μ s).
 - ▣ **InfiniBand:** Expensive but high performance (2 μ s latency, 200 Gbps).
- **Computation Example:**

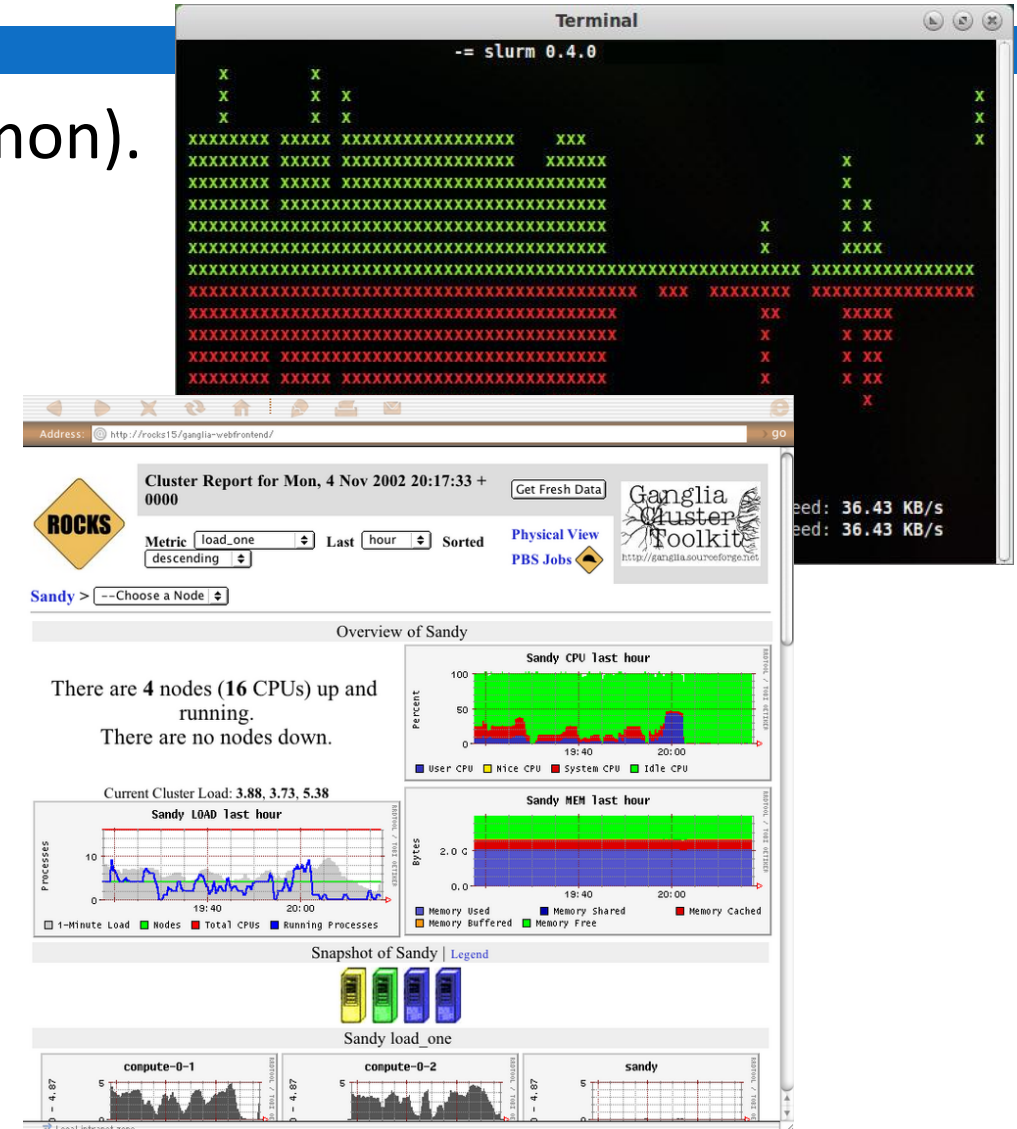
If each MPI message takes 5 μ s + 0.2 μ s per byte, estimate total latency for a 4 KB message.
- $L = 5 + 0.2 \times 4096 = 824.2 \mu$ s



Software Stack in Cluster Systems

15

- ❑ **Operating System:** Linux/Unix (most common).
- ❑ **Middleware Layer:**
 - ▣ MPI (Message Passing Interface)
 - ▣ PVM (Parallel Virtual Machine)
 - ▣ OpenMP (for hybrid models)
- ❑ **Resource Manager:**
 - ▣ SLURM, PBS, Torque
- ❑ **Monitoring Tools:**
 - ▣ Ganglia, Nagios, Prometheus
- ❑ **File System:**
 - ▣ NFS, Lustre, or GPFS for shared access.



Message Passing (MPI) Overview

16

- Each node runs a process; processes communicate via explicit message passing.
- Core MPI Functions:
 - `MPI_Init()`, `MPI_Send()`, `MPI_Recv()`, `MPI_Bcast()`, `MPI_Reduce()`.
- Example – Parallel Vector Addition:

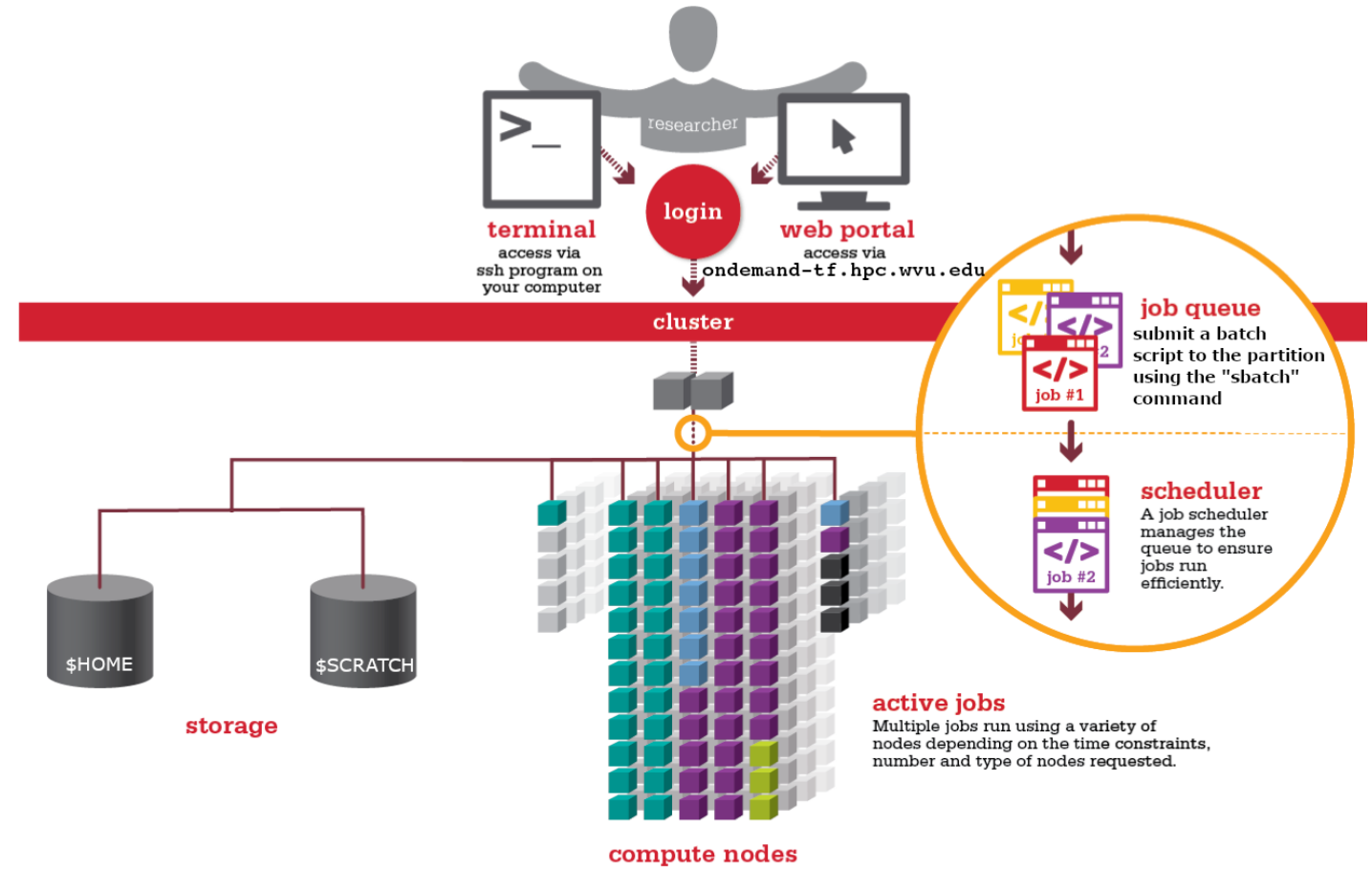
Each node adds its segment of the vector and sends results back to the master node.
- **Computation:**

If $N=1,000,000$ elements and 4 nodes, each processes 250,000 elements → **4× speedup** ideally.

Cluster Scheduling

17

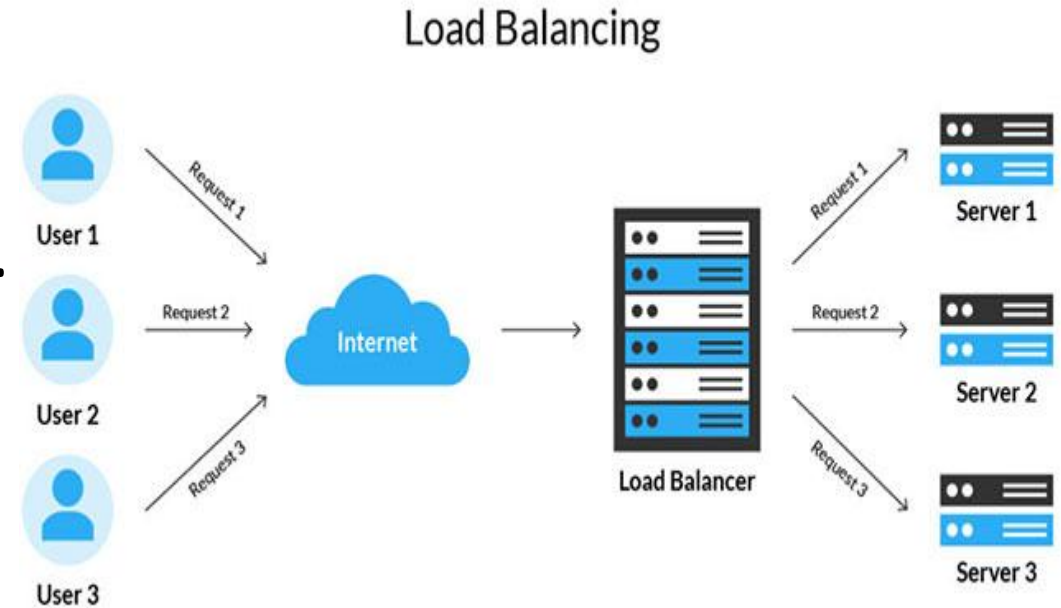
- ❑ **Schedulers:** SLURM, PBS, Sun Grid Engine.
- ❑ **Functions:**
 - ▣ Queue management.
 - ▣ Resource allocation.
 - ▣ Job prioritization.
- ❑ **Scheduling Policies:**
 - ▣ FIFO, Fair-share, Round-robin, Backfilling.



Load Balancing

18

- Balancing computational workload to prevent idle nodes.
 - **Static:** Equal division of tasks before execution.
 - **Dynamic:** Tasks redistributed at runtime.
 - **Strategies:**
 - ▣ Round Robin
 - ▣ Least Loaded Node
 - ▣ Work Stealing
- Example:** In heterogeneous clusters, tasks assigned proportional to CPU speed.

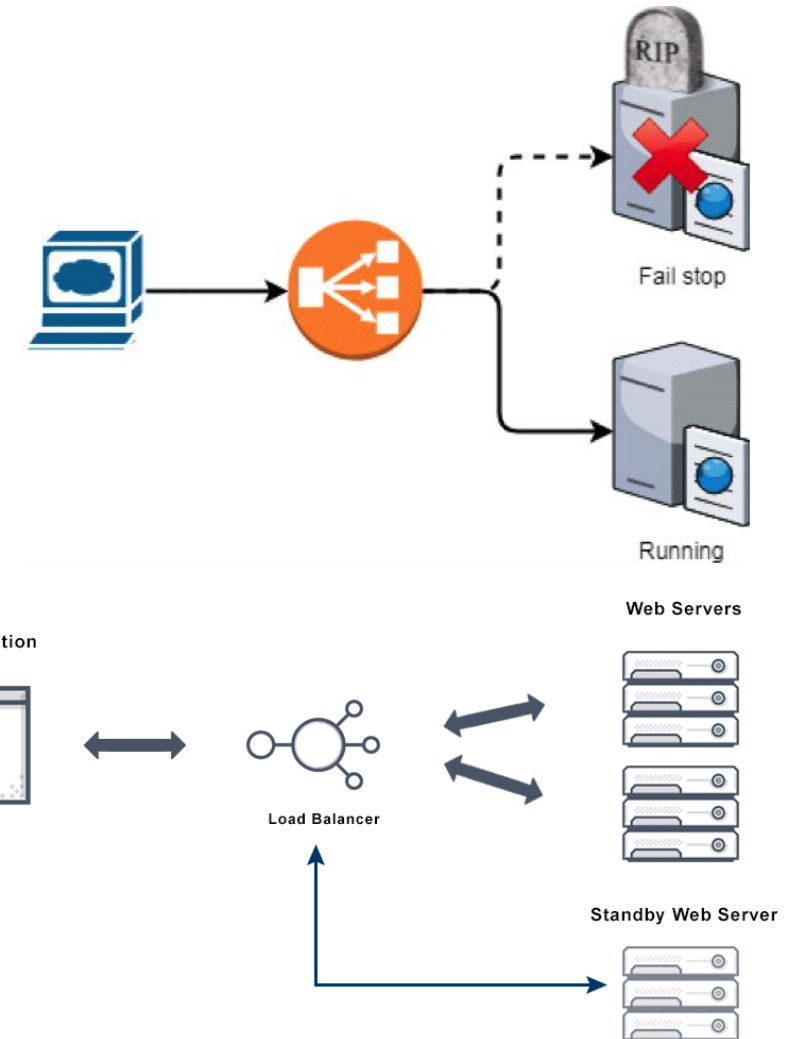


Fault Tolerance

19

- ❑ **Failure Types:** Node crash, network partition, disk failure.
- ❑ **Recovery Techniques:**
 - ▣ Checkpointing: Save progress periodically.
 - ▣ Redundancy: Replicate data or tasks.
 - ▣ Heartbeat Monitoring: Detect failed nodes.
- ❑ **Example:**

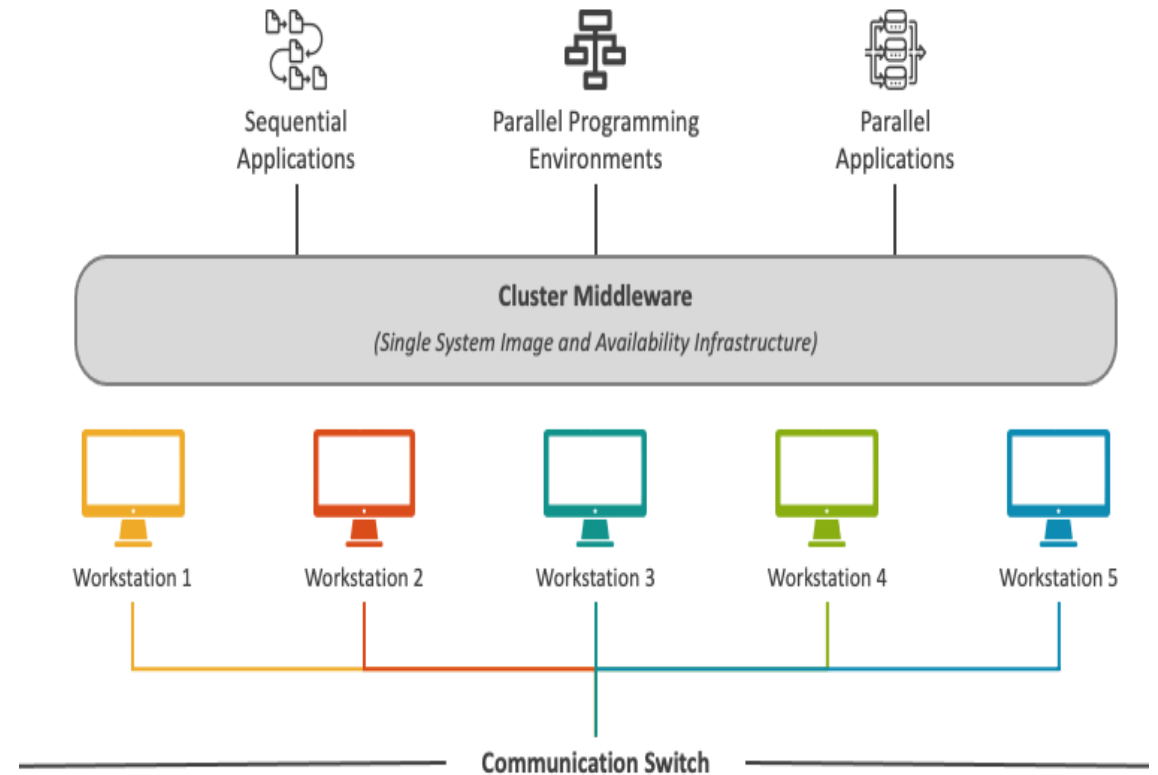
An MPI application can checkpoint every 10 minutes to resume after node failure.



Cluster Middleware

20

- ❑ Middleware = glue between OS and applications.
- ❑ Handles communication, resource management, and security.
- ❑ Examples:
 - ▣ **Rocks Cluster** (automated installation)
 - ▣ **OpenHPC** (open-source modular stack)
 - ▣ **Ganglia** (monitoring)
- ❑ Provides user-friendly command-line tools for launching jobs.



Case Study Calculation

21

- A model training takes 60 hours on 1 GPU node.
A 32-node cluster achieves **90% parallel efficiency**.

$$T_{cluster} = \frac{T_1}{N \times E} = \frac{60}{32 \times 0.9} = 2.08 \text{ hours}$$

- **Discussion:**

Is it worth using more nodes if efficiency drops below 70%?

Group Activity

22

- **Activity:**
- Form groups of 3–4.
- Choose one of these domains: Weather, Genomics, AI, or Rendering.
- Design a **mini-cluster setup** (5–10 nodes).
 - ▣ Specify node specs, interconnect type, scheduling software.
 - ▣ Estimate cost and expected performance.
- Present your plan next class in 3 minutes.

Summary & Reading

23

- ❑ Clusters are cost-effective alternatives to supercomputers.
- ❑ Rely on **distributed memory**, **message passing**, and **network interconnects**.
- ❑ Key design aspects: interconnect speed, load balancing, and scalability.
- ❑ Software tools like **MPI**, **SLURM**, and **Ganglia** are critical.
- ❑ Used widely in HPC, AI, and big data analytics.
- ❑ **Reading:** Pacheco, Ch. 3 – *Clusters and Networked Systems*

Thanks

24



**"SOME BEAUTIFUL
PATHS CAN'T BE
DISCOVERED WITHOUT
GETTING LOST."**

EROL OZAN

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